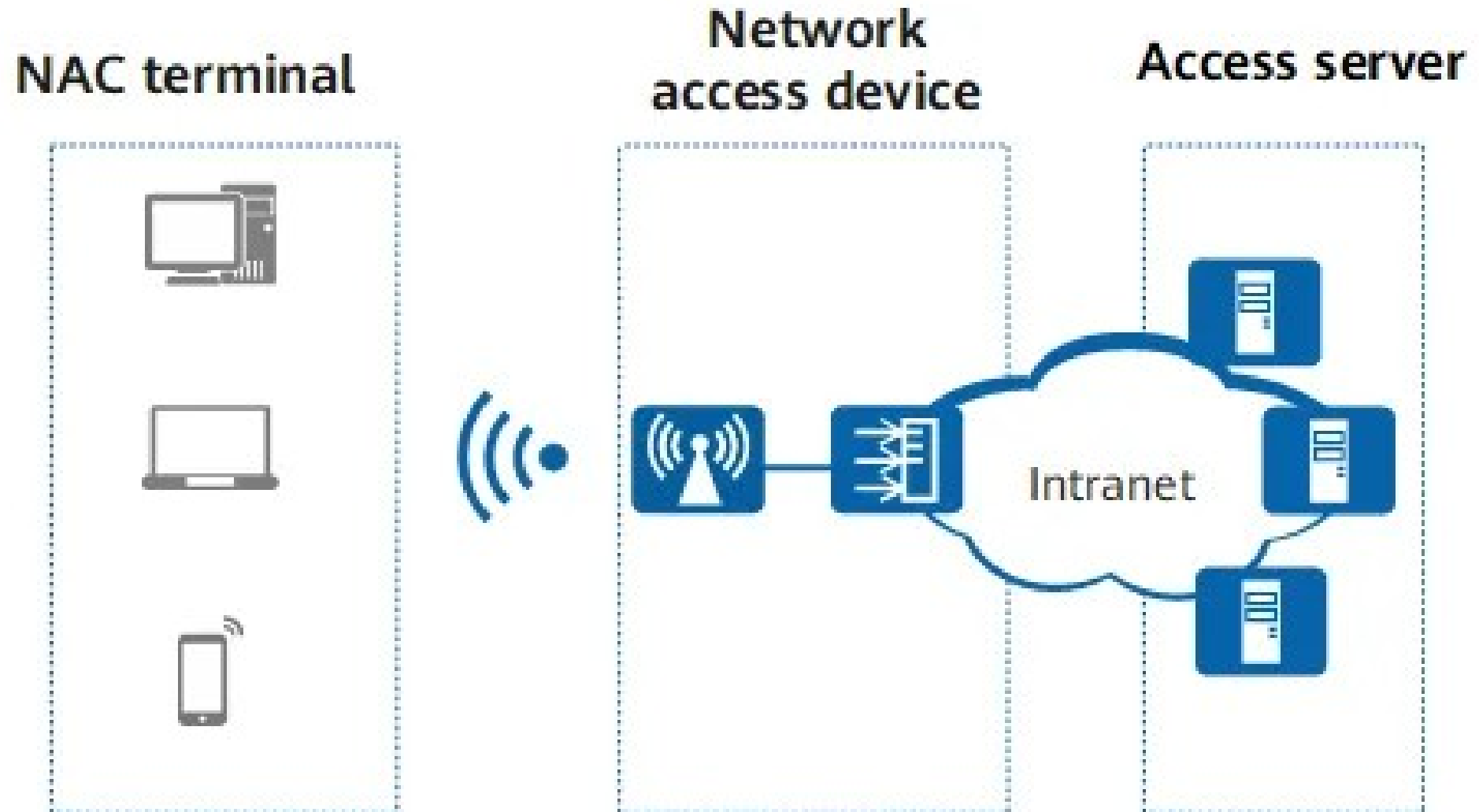


Network Access Control

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- Network Access Control is a security solution that uses a set of protocols to keep unauthorized users and devices out of a private network or give restricted access to devices that are compliant with network security policies.
- It handles network management and security that implements security policy, compliance, and management of access control to a network
- NAC works on wired and wireless networks by identifying different devices that are connected to the network.
- For setting up an NAC network security solution, administrators will determine the protocols that will decide how devices and users are authorized for the right level of authorization.
- Access rules are generally based on criteria such as the device used, the location accessed from, the access rights of various individuals, as well as the specific data and resources being accessed.

NAC



Components of Network Access Control Scheme:

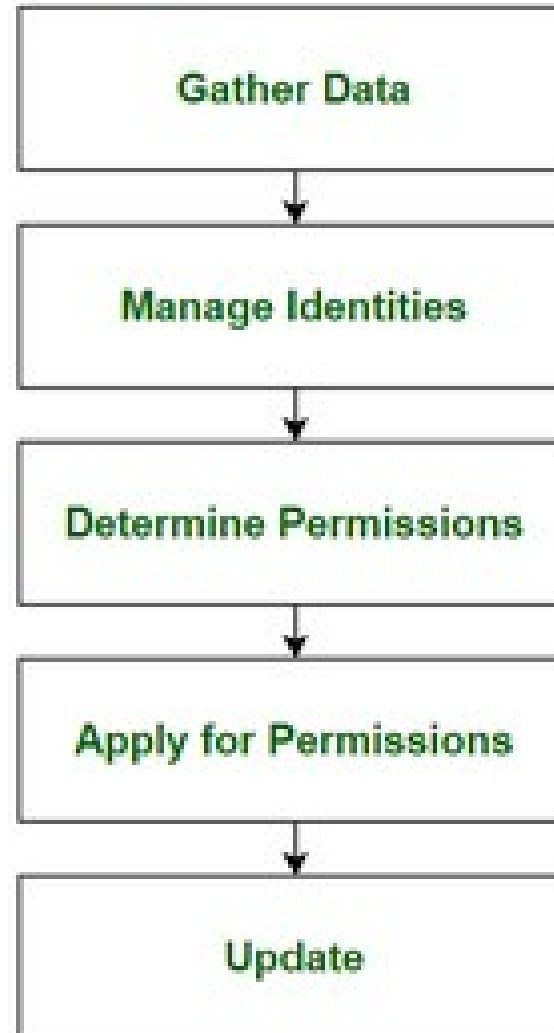
- 1. Restricted Access:** It restricts access to the network by user authentication and authorization control. For example, the user can't access a protected network resource without permission to access it.
- 2. Network Boundary Protection:** It monitors and controls the connectivity of networks with external networks. It includes tools such as controlled interfaces, intrusion detection, and anti-virus tools. It is also called perimeter defense. For example, the firewall can be used to prevent unauthorized access to network resources from outside of the network.

Types of Network Access Control:

- 1.Pre-admission:** It happens before access to the network is granted on initialization of request by the user or device to access the network. It evaluates the access attempt and only allows access if the user or device is compliant with organization security policies and authorized to access the network.
- 2.Post-admission:** It happens within the network when the user or device attempts to access the different parts of the network. It restricts the lateral movement of the device within the network by asking for re-authentication for each request to access a different part of the network.

Steps to Implement NAC Solutions:

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Steps to Implement NAC Solutions:

- 1.Gather Data:** Perform an exhaustive survey and collect information about every device, user, and server that has to interface with the network resources.
- 2.Manage Identities:** Verify user identities within the organization by authentication and authorization.
- 3.Determine Permissions:** Create permission policies stating different access levels for identified user groups.
- 4.Apply for Permissions:** Apply permission policies on identified user groups and register each user in the NAC system to trace their access level and activity within the network.
- 5.Update:** Monitor security operations and make adjustments to permission policies based on changing requirements of the organization with time.

Importance of Network Access Control:

- There has been exponential growth in the number of mobile devices accessing private networks of organizations in the past few years. This has led to an increase in security risks for the organization's resources and therefore, some tools are required that can provide the visibility, access control, and compliance capabilities to strengthen the network security infrastructure.
- A NAC system can deny network access to non-compliant devices or give them only restricted access to computing resources, thus preventing insecure nodes from infecting the network. Also, NAC products can handle large enterprise networks that have a large range of different device types connected to the network.

Responsibilities of NAC:

- 1.It allows only compliant, authenticated devices to access network resources and infrastructure.
- 2.It controls and monitors the activity of connected devices on the network.
- 3.It restricts the availability of network resources of private organizations to devices that follow their security policy.
- 4.It regulates the access of network resources to the users.
- 5.It mitigates network threats by enforcing security policies that block, isolate, and repair non-compliant machines without administrator attention.

Common Use-Cases:

- Organizations that allow employees to use their own devices or take corporate devices home use NAC to ensure network security.
- Organizations use NAC to grant access to different network resources to people or devices that are outside of the organization and are subject to different security controls.
- NAC protects from threats caused due to use of IoT devices by categorizing IoT devices into groups that have limited permission and constantly monitoring their activities.

Common Use-Cases:

- **IoT:** The use of IoT devices only continues to grow. This includes their use in Operational Technology (OT) settings and connections to enterprise networks from home networks. The right NAC solution will identify and monitor IoT devices, in addition to traditional devices.
- **Incident Response:** In addition to simply controlling network access, a robust network access control solution should be able to respond to threats quickly and effectively. This is where automation comes into play. Automation in a NAC solution enforces security policies, shares contextual information, and isolates insecure devices at the point of connection to the network before they can do any damage

Common Use-Cases:

- **Contractors:** Often, companies want to allow contractors, partners, or temporary workers access to only certain parts of the network. NAC can be used to maintain access privileges and prevent unauthorized access to certain parts of the network while ensuring guest users have smooth connectivity and a good experience.
- **Medicine:** In the world of healthcare, there is a growing reliance on the Internet of Medical Things (IoMT) devices. But healthcare is a highly regulated industry, and network compliance is vital. Properly structured NAC solutions can provide the necessary protection of sensitive personal data and medical records in a network with multiple users and IoMT devices.

Benefits:

1. Users can be required to authenticate via multi-factor authentication, which is much more secure than identifying users based on IP addresses or username and password combinations.
2. It provides additional levels of protection around individual parts of the network.

Limitations:

- 1.It has low visibility in IoT devices and devices with no specific users associated with it.
- 2.It does not protect from threats present inside the network.
- 3.It may not work for organizations if it is not compatible with existing security controls.

Principle Components of NAC(Network Access Control):

There are mainly three principle elements of NAC which are:

1. Access Requestor(AR).
2. Policy Servers.
3. Network Access Servers(NAS).
4. User and Device Identification Mechanisms
5. Network Admission Control
6. Quarantine and Guest Network Management

Access Requestor(AR):

- We may determine from the name that it is someone attempting to gain access by requesting it. This access can be granted to any entity, such as a device, person, or process.
- This entity attempts to get access to network resources. It might be any device handled by the NAC system, such as servers, cameras, printers, and other IP-enabled devices.
- ARs ensures that no entity has illegal access to protected resources.
- To get access, these ARs must follow the organization's specific guidelines or policies.

Policy Server:

- The policy server analyzes what access should be provided to AR based on the AR's identity, permission level, attempted request, and an organization's established access policy.
- The policy server frequently relies on backend services, such as antivirus, patch management, or a user directory, to function.
- The policy server helps to determine the host's state. An organization creates different access policies to clearly authorize or reject such access. If the AR follows the organization's policy, the policy server gives access based on the requestor's permission; otherwise, the AR will not be permitted access based on its permission.
- It should be noted that there are various commercial systems on the market now that provide such policy servers for both on-premises and cloud computing. Some of the most common examples include the Cisco Identity Services Engine(ISE), Forescout Platform, Aruba ClearPass Policy Manager, and FortiNAC.
- These tools offer highly detailed ways to set organizational rules and control the organization's full IP infrastructure.

Network Access Server(NAS):

- Users connecting to an organization's internal network from distant locations utilize the NAS as an access control point. These often serve as VPNs and give users access to the company's internal network. These days, NAS functionality is frequently included in policy server systems.
- Remote employees can connect to the company's internal network via NAS, which serves as an access point for them. This allows the company and its employees to create a secure connection and grant authorized access to the network.

User and Device Identification Mechanisms

- Effective identification mechanisms are essential for a NAC system.
- These mechanisms involve various methods such as usernames and passwords, digital certificates, or multi-factor authentication.
- By accurately identifying users and devices, the NAC system can apply the appropriate access policies and ensure that only authorized entities gain network access.

Network Admission Control

- Network Admission Control is a specialized subset of NAC focused on evaluating device compliance before granting network access.
- This involves conducting rigorous security posture checks, such as verifying antivirus updates and security patches, to ensure devices meet organizational standards.
- Devices that fail to meet these criteria may be quarantined or given limited access until necessary updates are applied.

Quarantine and Guest Network Management

- Managing non-compliant devices and guest access is a critical function of a NAC system.
- Quarantine networks isolate devices that fail to meet security standards, guiding users through remediation steps to achieve compliance.
- Guest network management provides temporary, restricted access to visitors, ensuring they can connect without compromising sensitive internal resources.
- This dual approach maintains network integrity while accommodating diverse access needs.

NAC Enforcement Methods

- Network Access Control (NAC) enforces security policies on devices seeking to access a network. There are several NAC enforcement methods:
 - **In-line Enforcement:** This method places NAC devices in-line with network traffic, allowing or denying access based on security policy compliance.
 - **Out-of-Band Enforcement:** Here, NAC devices monitor network traffic and communicate with other network devices to enforce security policies without being directly in the traffic path.
 - **Agent-Based Enforcement:** This method involves installing software agents on devices to enforce security policies and ensure compliance before granting network access.
 - **Agentless Enforcement:** Unlike agent-based enforcement, this method does not require installing software agents on devices. Instead, it uses existing protocols and methods to assess and enforce security policies.
 - **802.1X Enforcement:** This method uses the IEEE 802.1X standard to control access to a network, requiring authentication before allowing devices to connect.

Each enforcement method has its advantages and limitations, and the choice of method depends on the specific security requirements and network infrastructure.

NAC Enforcement Methods

- **Fire-wall:** A firewall provides a form of NAC by allowing and denying network traffic between the host and the external user.
- **Device Authentication:** Verifying devices attempting to connect using credentials, certificates, or MAC address filtering.
- **User Role-Based Access Control:** Assigning access rights based on user roles, ensuring users have the minimum necessary permissions.
- **Posture Assessment:** Checking the security compliance of devices (e.g., antivirus, OS updates) before granting access.
- **Quarantine Networks:** Isolating non-compliant devices to a limited network until they meet security requirements.
- **Dynamic VLAN Assignment:** Assigning users to different Virtual Local Area Networks (VLANs) based on their identity or security posture.
- **Logging and Monitoring:** Continuously monitoring access logs and network activity for anomalies or unauthorized access attempts.
- **Policy Enforcement:** Implementing security policies that define acceptable use and access rights.
- **Remediation Actions:** Automatically initiating actions to remediate non-compliant devices (e.g., pushing updates or alerting IT staff).